CRYSS Monte Carlo + Optimizer Summary

Technical Overview for Future Users and Coders

CRYSS Project

October 18, 2026

1 Purpose

This note summarizes the current optimization strategy used in the CRYSS workflow and explains why the project now uses a Monte Carlo (MC) search followed by a local refinement step.

The key point is that the system does not know the final maximum recovery $R_{\max}$ in advance. Instead, it observes composition data at known slice recoveries R_s and fits the mechanistic model so that the predicted solid compositions match the measured slice compositions. The optimizer therefore operates on the observed slice data, not on an assumed or directly measured $R_{\max}$.

2 Governing Principle

For each mixture, the model is evaluated at the known slice recoveries $R_1, R_2, \ldots, R_n$. At each slice, the model produces a predicted solid-phase composition vector

$$\mathbf{X}^{\text{pred}}(R_s; T_m, \Delta H) \tag{1}$$

and the project compares this against the measured composition vector

$$\mathbf{X}^{\text{obs}}(R_s). \tag{2}$$

The optimization target is therefore a direct slice-fit objective:

$$J(T_m, \Delta H) = \sum_{s=1}^{n} w_s \sum_{i=1}^{m} \left(X_i^{\text{pred}}(R_s; T_m, \Delta H) - X_i^{\text{obs}}(R_s) \right)^2, \tag{3}$$

where m is the number of components, n is the number of slice measurements, and w_s are optional slice weights.

This is intentionally different from a direct optimization of $R_{\max}$, because $R_{\max}$ is inferred from the model after fitting the slice-wise composition data.

3 Why Monte Carlo Comes First

The thermodynamic parameter space is not smooth enough and not sufficiently well constrained for a direct single-pass optimizer to be reliable across all mixtures. In particular:

- the problem is nonlinear,
- the parameter landscape is multi-modal,
- several mixtures exhibit difficult or outlying behaviour,
- the objective is tied to slice-wise composition matching rather than a directly known scalar target.

For this reason, the production workflow uses a stochastic broad search to find a good basin of attraction before local refinement. The Monte Carlo stage explores many candidate $(T_m, \Delta H)$ states and ranks them by the direct slice-fit objective.

4 MC Candidate Pool and Local Refinement

The current workflow is:

1. Generate a Monte Carlo candidate pool.
2. Rank the candidates by the direct slice-fit objective.
3. Retain the best k candidates for diagnostic and comparison purposes.
4. Refine the best candidate, or the best few candidates, using a local bounded search around the seed value.

In the present implementation, the local optimization is intentionally seeded from the best MC result or from the best few MC results, depending on the analysis mode. The comparison study used the best 10 MC candidates as the seed set to test whether the optimizer can effectively remove difficult outliers and tighten the final prediction. An additional point, during the Monte Carlo search it was observed that the Halley solver became unstable under PD mode conditions due to use of stretching input parameters in the somewhat complex residual. Thus, in the cases where a system bears a PD mode flag, the code-flow switches to the VBA solver trading a speed for stability.

5 Objective Design

The project deliberately separates the following concerns:

- **MC objective**: direct fit to observed slice compositions at known slice recoveries.
- **Optimizer objective**: same direct slice-fit objective, used for local refinement.
- **Tensor objective**: This is a legacy solution which turned out not to be necessary; it is present in the code base but not actively used in the current workflow.

6 Observed Result from the 50-Mixture Study

The comparison run over 50 unique mixtures produced a strong result. The best MC candidates and the MC+optimizer candidates were plotted against the NODES reference values. The fitted relationship is close to the identity line, with a very strong coefficient of determination, approximately

$$R^2 \approx 0.994. \tag{4}$$

This is a strong indication that the optimizer is not simply redistributing noise. Instead, it is reducing the systematic deviation and tightening the fit around the reference values.

The practical effect observed in the study was that the optimizer successfully removed the trickier outlier behaviour and produced a much cleaner final recovery relationship when seeded from the best MC candidates.

7 Interpretation

This indicates that the project has reached a useful and stable operational pattern:

- MC provides the broad exploratory search,
- MC ranking identifies promising basin candidates,
- local optimization refines the strongest basin,
- the final result is strongly aligned with the reference data.

In other words, the workflow is doing the correct kind of work: it is not trying to fit a hidden target directly, but rather fitting the model to the measured slice data, then projecting that model to the recovery behaviour implied by the slice-wise composition match.

8 Recommended Production Flow

The recommended production pattern is:

1. Use Monte Carlo to generate a broad candidate set.
2. Rank candidates by the direct slice-fit objective.
3. Retain the best $10-20$ candidates for comparison and diagnostics.
4. Refine the best MC seed or best few MC seeds with a bounded local search.
5. Use the optimized candidate as the final fit for the mixture.

This gives a robust decomposition between search and refinement, while keeping the objective grounded in the actual observed slice data.

9 Summary

The main lesson is simple: the model is fit from the slice data, not from an assumed direct measurement of $R_{\max}$. The MC+optimizer workflow is therefore best understood as a two-stage process:

- broad search over physically plausible parameter values,
- local refinement against the measured slice compositions.

The observed performance on the 50-mixture comparison is strong evidence that this approach is effective and that the best MC candidates are the correct seeds for the local optimizer.